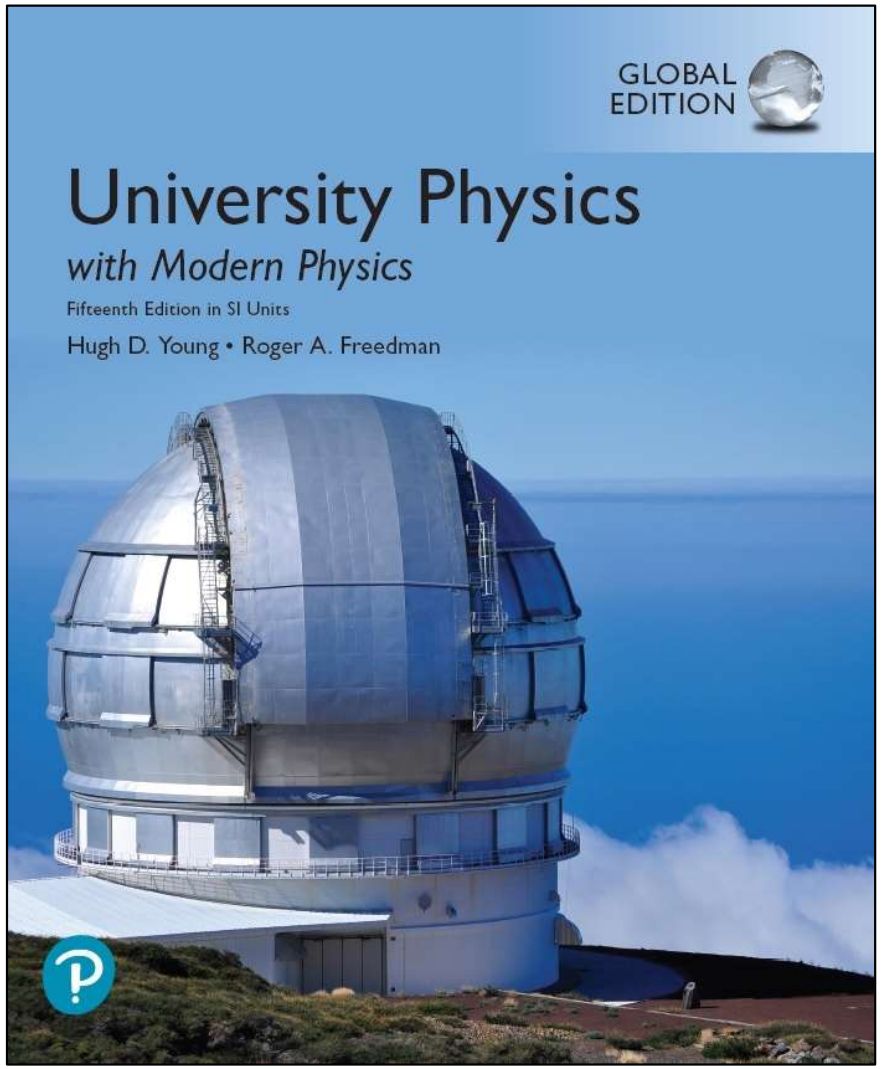
University Physics with Modern Physics

Fifteenth Edition, Global Edition, in SI Units



Chapter 32

Electromagnetic Waves

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Learning Outcomes

In this chapter, you'll learn…

- how electromagnetic waves are generated.
- how and why the speed of light is related to the fundamental constants of electricity and magnetism.
- how to describe the propagation of a sinusoidal electromagnetic wave.
- what determines the amount of energy and momentum carried by an electromagnetic wave.
- how to describe standing electromagnetic waves.

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Introduction

- Why do metals reflect light?
- We will see that light is an electromagnetic wave.
- There are many other examples of electromagnetic waves, such as radiowaves and x rays.



- Unlike sound or waves on a string, these waves do not require a medium to travel.

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James Clerk Maxwell and Electromagnetic Waves

- The Scottish physicist James Clerk Maxwell (1831–1879) was the first person to truly understand the fundamental nature of light.
- He proved in 1865 that an electromagnetic disturbance should propagate in free space with a speed equal to that of light.
- From this, he deduced correctly that light was an electromagnetic wave.



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Electricity, Magnetism, and Light

- According to Maxwell's equations, an accelerating electric charge must produce electromagnetic waves.
- For example, power lines carry a strong alternating current, which means that a substantial amount of charge is accelerating back and forth and generating electromagnetic waves.
- These waves can produce a buzzing sound from your car radio when you drive near the lines.

(b)

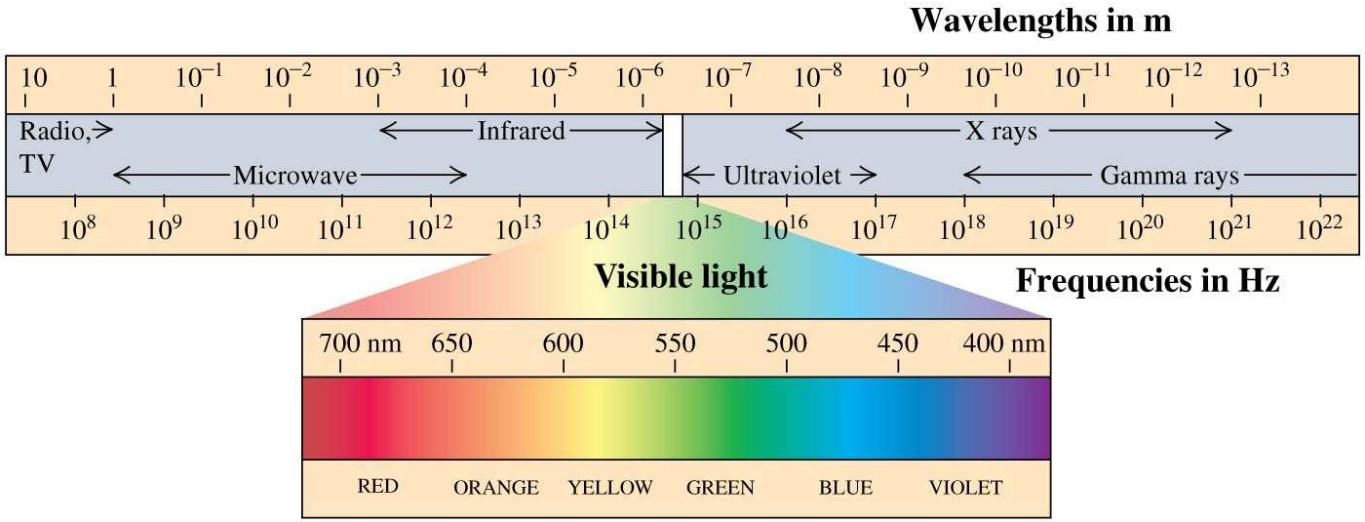


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The Electromagnetic Spectrum

- The frequencies and wavelengths of electromagnetic waves found in nature extend over such a wide range that we have to use a logarithmic scale to show all important bands.
- The boundaries between bands are somewhat arbitrary.



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Visible Light

- **Visible light** is the segment of the electromagnetic spectrum that we can see.
- Visible light extends from the violet end (400 nm) to the red end (700 nm).

Table 32.1 Wavelengths of Visible Light

380–450 nm	Violet
450–495 nm	Blue
495–570 nm	Green
570–590 nm	Yellow
590–620 nm	Orange
620–750 nm	Red

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Ultraviolet Vision

- Many insects and birds can see ultraviolet wavelengths that humans cannot.
- As an example, the left-hand photo shows how black-eyed Susans look to us.



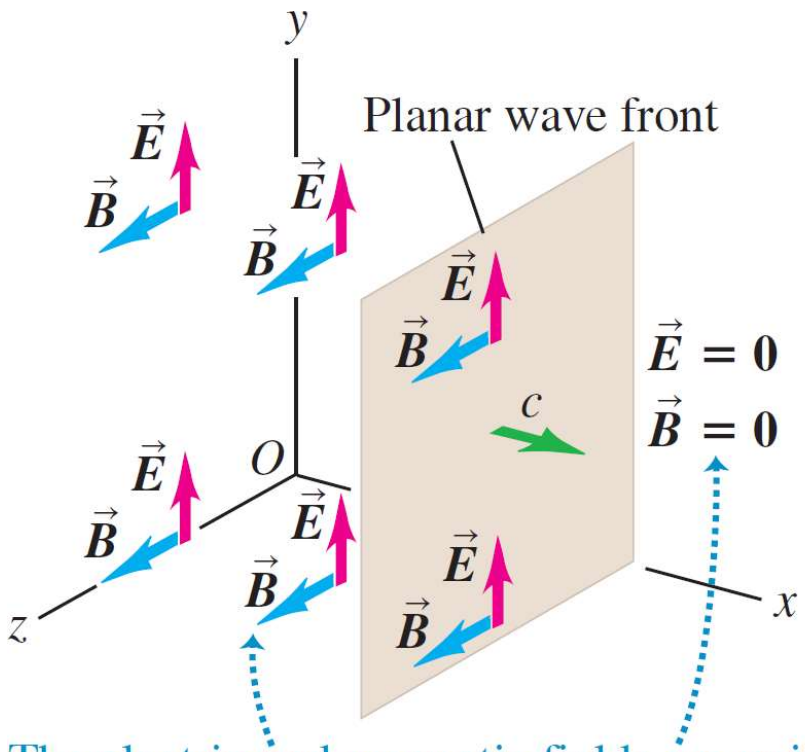
- The right-hand photo (in false color), taken with an ultraviolet-sensitive camera, shows how these same flowers appear to the bees that pollinate them.
- Note the prominent central spot that is invisible to humans.

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A Simple Plane Electromagnetic Wave

- To begin our study of electromagnetic waves, imagine that all space is divided into two regions by a plane perpendicular to the *x*-axis.
- At every point to the left of this plane there are uniform electric field magnetic fields as shown.
- The boundary plane, which we call the **wave front**, moves in the +*x*-direction with a constant speed *c*.



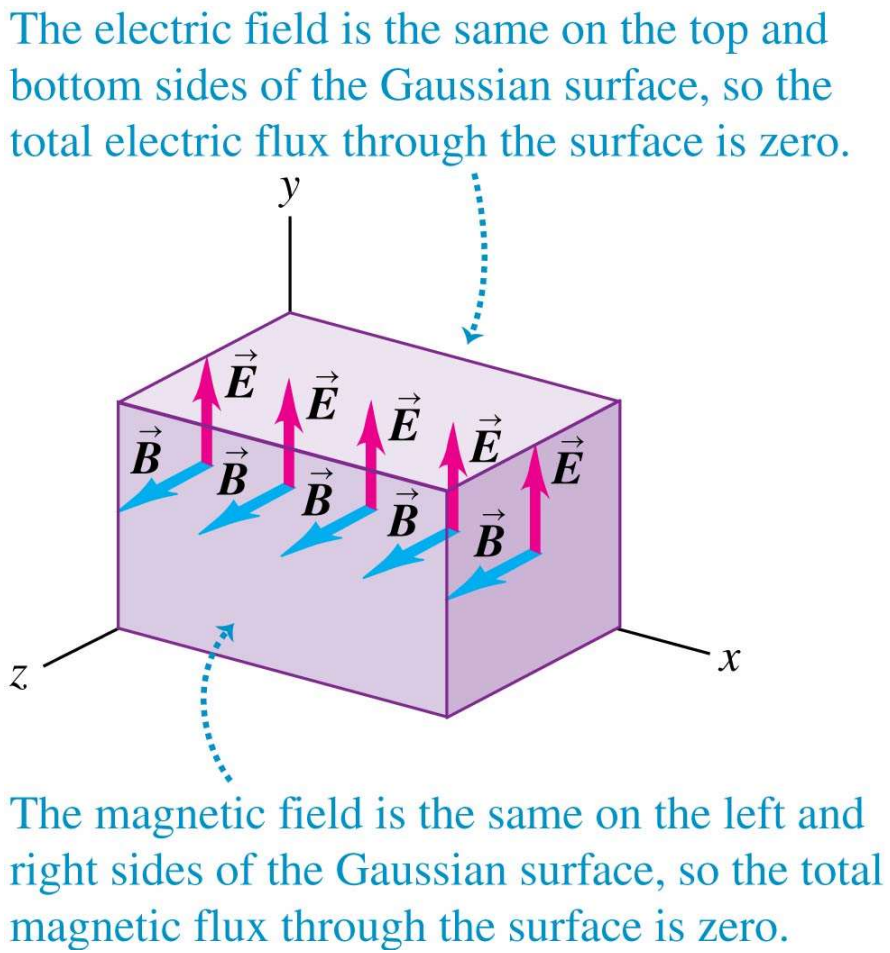
The electric and magnetic fields are uniform behind the advancing wave front and zero in front of it.

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Gauss's Laws and the Simple Plane Wave

- Shown is a Gaussian surface, a rectangular box, through which the simple plane wave is traveling.
- The box encloses no electric charge.
- In order to satisfy Maxwell's first and second equations, the electric and magnetic fields must be perpendicular to the direction of propagation; that is, the wave must be **transverse**.



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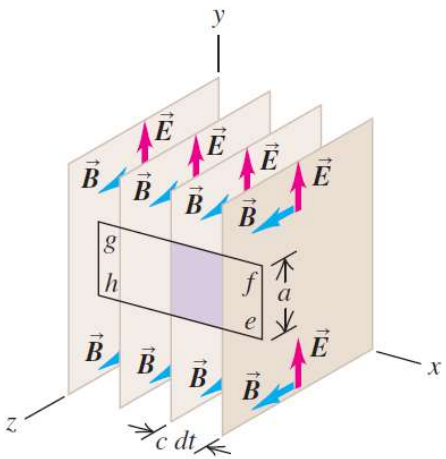
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Faraday's Law and the Simple Plane Wave

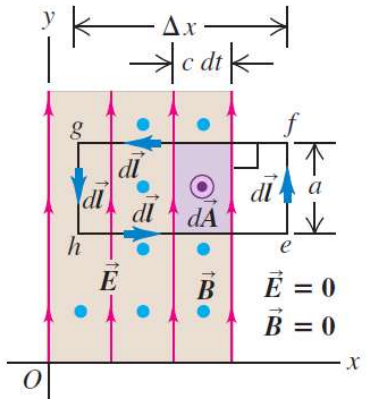
- The simple plane wave must satisfy Faraday's law in a vacuum.
- In a time *dt*, the magnetic flux through the rectangle in the *xy*-plane increases by an amount $d\Phi_B$.
- This increase equals the flux through the shaded rectangle with area *ac dt*; that is, $d\Phi_B = Bac\,dt$.
- Thus $d\Phi_B / dt = Bac$.
- This and Faraday's law imply:

Electromagnetic wave in vacuum: $E = cB$ (Electric-field magnitude; Magnetic-field magnitude; Speed of light in vacuum)

(a) In time *dt*, the wave front moves a distance *c dt* in the +*x*-direction.



(b) Side view of situation in (a)



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Ampere's Law and the Simple Plane Wave

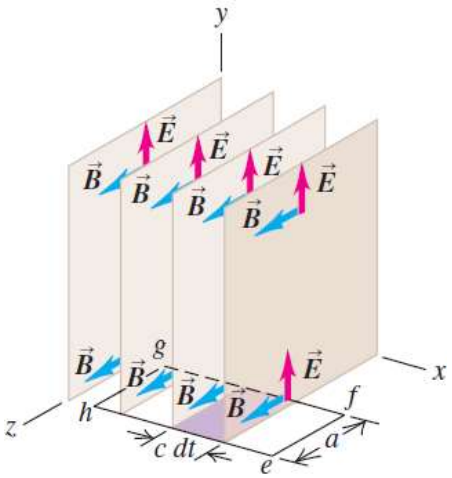
- The simple plane wave must satisfy Ampere's law in a vacuum. In a time *dt*, the electric flux through the rectangle in the *xz*-plane increases by an amount $d\Phi_E$.
- This increase equals the flux through the shaded rectangle with area *ac dt*; that is, $d\Phi_E = Eac\,dt$.
- Thus $d\Phi_E / dt = Eac$. This implies:

Electromagnetic wave in vacuum:

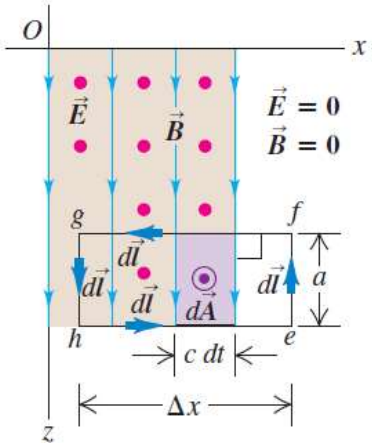
$$B = \epsilon_0 \mu_0 c E$$

Magnetic-field magnitude; Electric-field magnitude; Speed of light in vacuum; Electric constant; Magnetic constant

(a) In time *dt*, the wave front moves a distance *c dt* in the +*x*-direction.



(b) Top view of situation in (a)



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Properties of Electromagnetic Waves (1 of 2)

- Maxwell's equations imply that in an electromagnetic wave, both the electric and magnetic fields are perpendicular to the direction of propagation of the wave, and to each other.
- In an electromagnetic wave, there is a definite ratio between the magnitudes of the electric and magnetic fields: $E = cB$.
- Unlike mechanical waves, electromagnetic waves require no medium. In fact, they travel in vacuum with a definite and unchanging speed:

Speed of electromagnetic waves in vacuum $c = \dfrac{1}{\sqrt{\epsilon_0 \mu_0}}$ (Electric constant, Magnetic constant)

- Inserting the numerical values of these constants, we obtain $c = 3.00 \times 10^8$ m/s.

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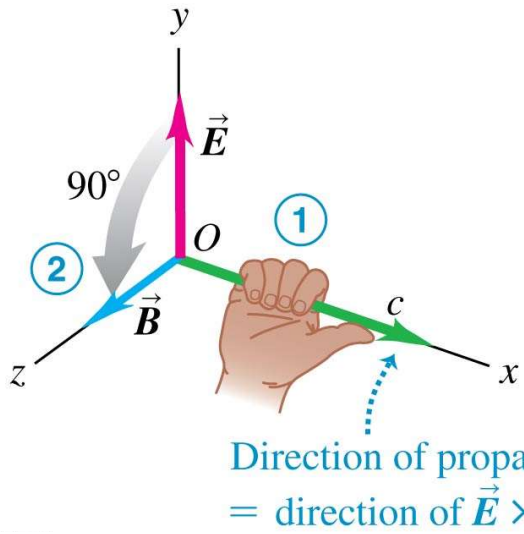
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Properties of Electromagnetic Waves (2 of 2)

- The direction of propagation of an electromagnetic wave is the direction of the vector product of the electric and magnetic fields.

Right-hand rule for an electromagnetic wave:

1. Point the thumb of your right hand in the wave's direction of propagation.
2. Imagine rotating the $\vec{E}$-field vector 90° in the sense your fingers curl. That is the direction of the $\vec{B}$ field.

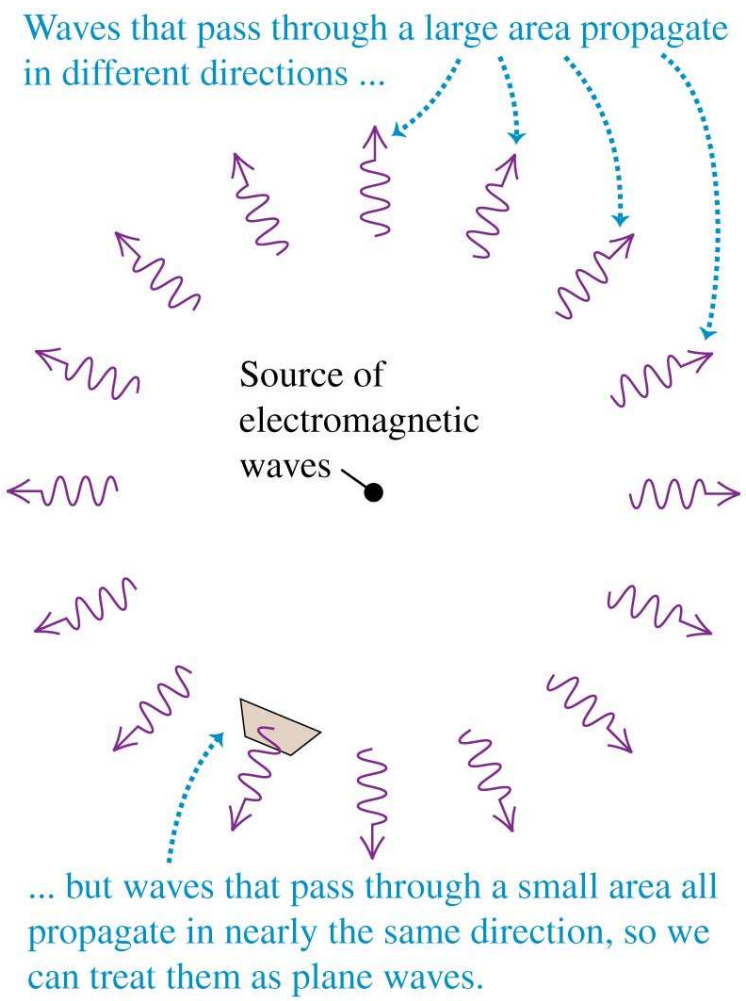


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Sinusoidal Electromagnetic Waves

- Electromagnetic waves produced by an oscillating point charge are an example of sinusoidal waves that are not plane waves.
- But if we restrict our observations to a relatively small region of space at a sufficiently great distance from the source, even these waves are well approximated by plane waves.

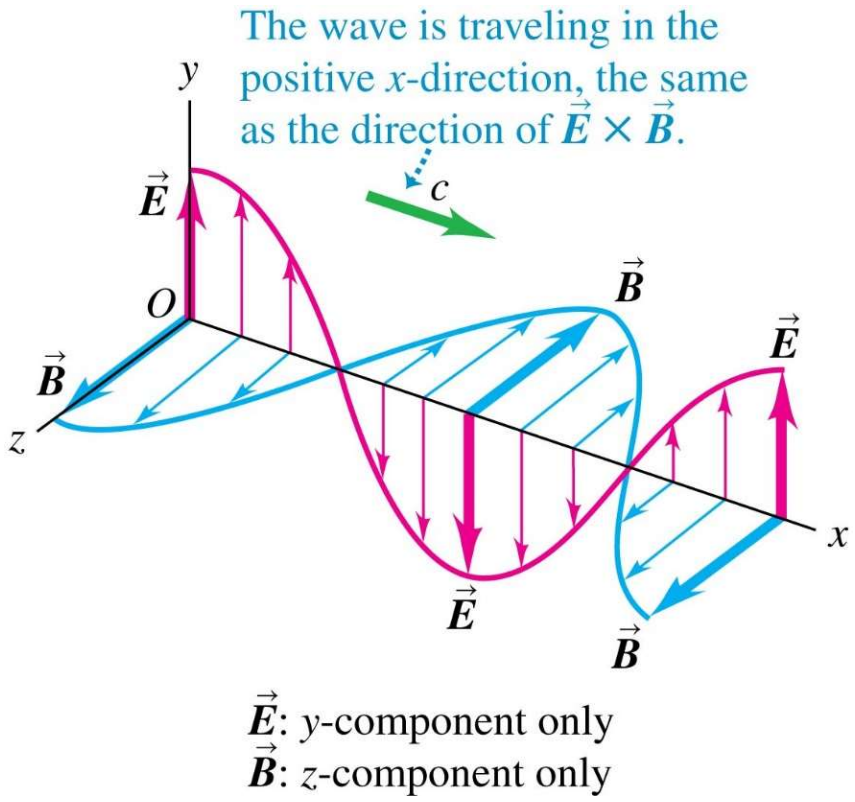


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Fields of a Sinusoidal Wave (1 of 3)

- Shown is a linearly polarized sinusoidal electromagnetic wave traveling in the +*x*-direction.
- One wavelength of the wave is shown at time $t = 0$.
- The fields are shown for only a few points along the *x*-axis.

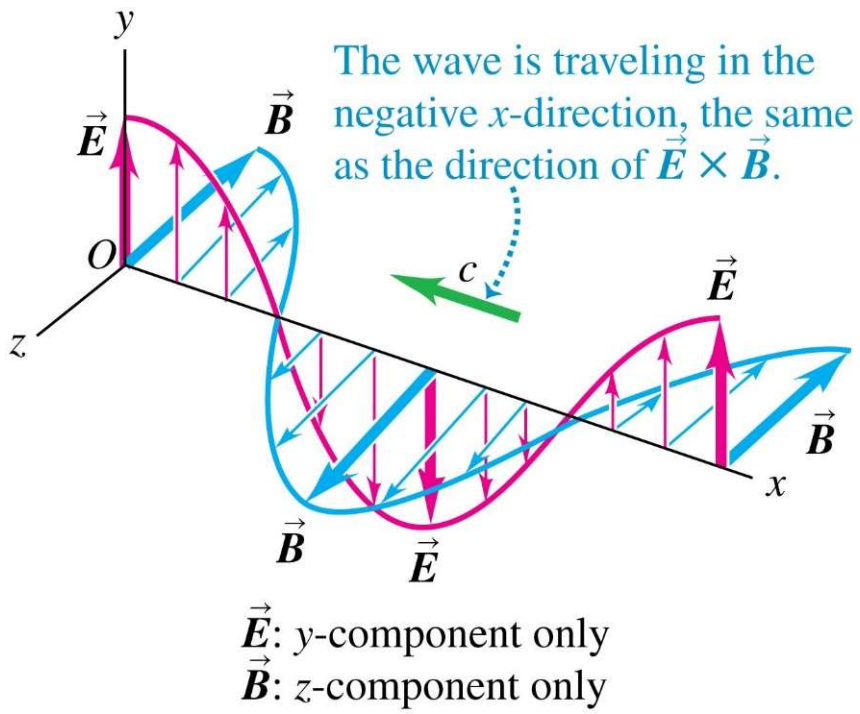


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Fields of a Sinusoidal Wave (2 of 3)

- Shown is a linearly polarized sinusoidal electromagnetic wave traveling in the −*x*-direction.
- One wavelength of the wave is shown at time $t = 0$.
- The fields are shown for only a few points along the *x*-axis.



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Fields of a Sinusoidal Wave(3 of 3)

- We can describe electromagnetic waves by means of wave functions:

Sinusoidal electromagnetic plane wave, propagating in $+x$-direction:

Electric field — Electric-field magnitude — Wave number — Angular frequency

$$\vec{E}(x, t) = \hat{j}E_{\max}\cos(kx - \omega t)$$

$$\vec{B}(x, t) = \hat{k}B_{\max}\cos(kx - \omega t)$$

Magnetic field — Magnetic-field magnitude

- The wave travels to the right with speed $c = \dfrac{\omega}{k}$.
- The amplitudes must be related by:

Sinusoidal electromagnetic wave in vacuum:

Electric-field amplitude — Magnetic-field amplitude — Speed of light in vacuum

$$E_{\max} = cB_{\max}$$

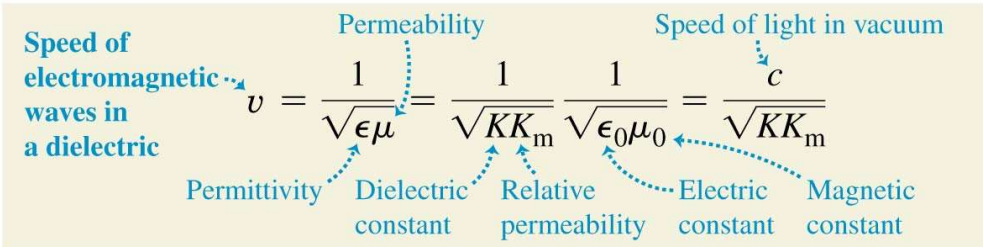
- Video Tutor Solution: Example 32.1

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Electromagnetic Waves in Matter

- Electromagnetic waves can travel in certain types of matter, such as air, water, or glass.
- When electromagnetic waves travel in nonconducting materials—that is, dielectrics—the speed v of the waves depends on the dielectric constant of the material.



- The ratio of the speed c in vacuum to the speed v in a material is known in optics as the **index of refraction** n of the material.

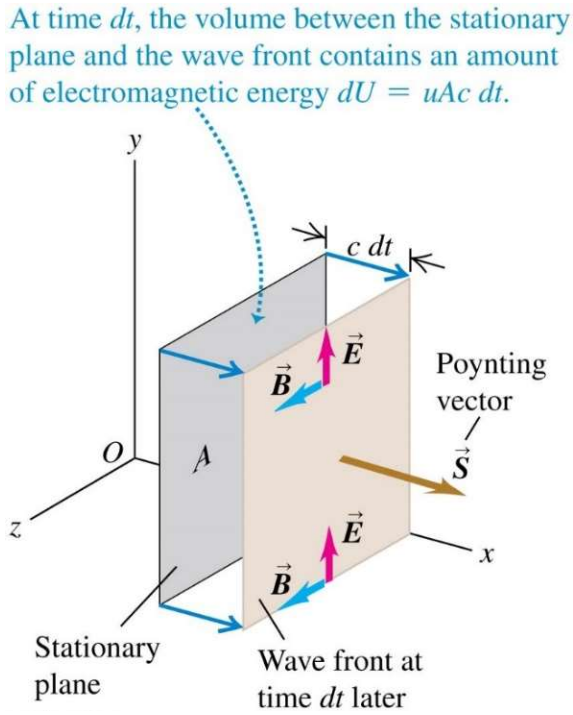
$$\frac{c}{v} = n = \sqrt{KK_m} \cong \sqrt{K}$$

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Energy in Electromagnetic Waves (1 of 2)

- Electromagnetic waves such as those we have described are traveling waves that transport energy from one region to another.
- The British physicist John Poynting introduced the **Poynting vector**, $\vec{S}$.
- The magnitude of the Poynting vector is the power per unit area in the wave, and it points in the direction of propagation.



Poynting vector in vacuum: $\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$ (Electric field; Magnetic field; Magnetic constant)

- Video Tutor Solution: Example 32.6

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Energy in Electromagnetic Waves (2 of 2)

- The magnitude of the average value of $\vec{S}$ is called the **intensity**. The SI unit of intensity is $1\,\mathrm{W/m^2}$.
- These rooftop solar panels are tilted to be face-on to the sun so that the panels can absorb the maximum amount of wave energy.



Intensity of a sinusoidal electromagnetic wave in vacuum

Electric-field amplitude; Magnetic-field amplitude; Electric constant

$$I = S_{\mathrm{av}} = \frac{E_{\mathrm{max}} B_{\mathrm{max}}}{2\mu_0} = \frac{E_{\mathrm{max}}^2}{2\mu_0 c} = \frac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}\,E_{\mathrm{max}}^2 = \frac{1}{2}\epsilon_0 c E_{\mathrm{max}}^2$$

Magnitude of average Poynting vector; Magnetic constant; Speed of light in vacuum

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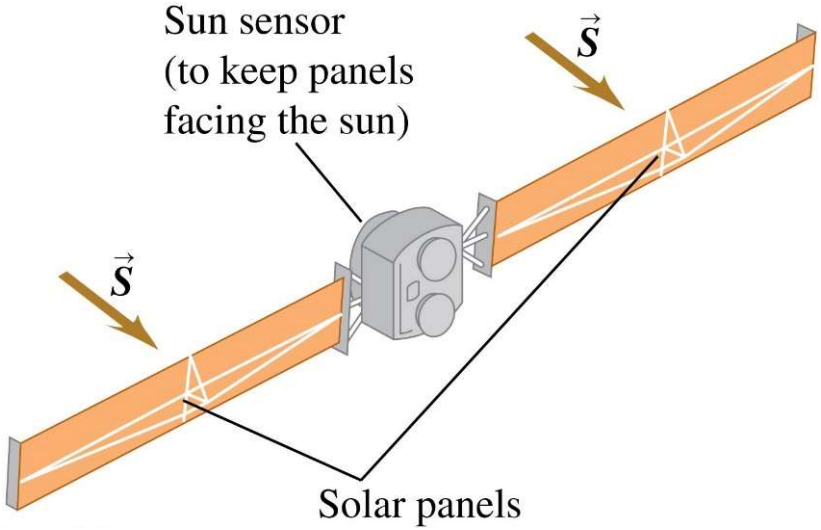
Electromagnetic Radiation Pressure

- Electromagnetic waves carry momentum and can therefore exert **radiation pressure** on a surface:

$$p_{\text{rad}} = \frac{S_{\text{av}}}{c} = \frac{I}{c} \quad \text{(radiation pressure, wave totally absorbed)}$$

$$p_{\text{rad}} = \frac{2S_{\text{av}}}{c} = \frac{2I}{c} \quad \text{(radiation pressure, wave totally absorbed)}$$

- For example, if the solar panels on an earth-orbiting satellite are perpendicular to the sunlight, and the radiation is completely absorbed, the average radiation pressure is 4.7×10^{-6} N/m^2.

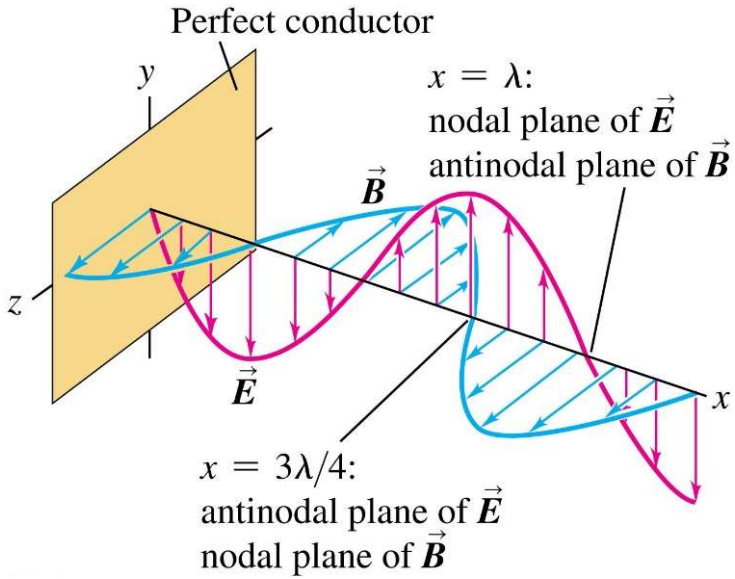


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Standing Electromagnetic Waves

- Electromagnetic waves can be reflected by a conductor or dielectric, which can lead to **standing waves**.
- As time elapses, the pattern does not move along the *x*-axis; instead, at every point the electric and magnetic field vectors simply oscillate.



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Standing Waves in a Cavity

- A typical microwave oven sets up a standing electromagnetic wave with λ = 12.2 cm, a wavelength that is strongly absorbed by the water in food.
- Because the wave has nodes spaced $\frac{\lambda}{2} = 6.1\,\text{cm}$ apart, the food must be rotated while cooking.
- Otherwise, the portion that lies at a node—where the electric-field amplitude is zero—will remain cold.
- Video Tutor Solution: Example 32.7



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